

Shubham Kesarwani

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PROFESSIONAL SUMMARY

Lead Engineer with 6+ years of experience in **Embedded Device Driver** and **Application Development**. Proven technical leader specializing in **C++, C#, .NET, Python, and Java**. Expert in **Driver Development** and **Tools Engineering** within the **IoT Smart Home** and **Healthcare** sectors. Successfully led cross-functional teams to deliver scalable integration tools bridging firmware and software layers. Strong expertise in **System Design, Agile methodologies**, and **SQL Server**, with a track record of mentoring engineers and delivering high-performance, mission-critical solutions

CORE SKILLS

- **Languages & Frameworks:** C++, Python, C#, .NET (WinForms, WPF, WCF), Java, SQL Server
- **Embedded Systems & IoT Protocols:** UART, RS232, Ethernet, BLE (802.15), Wi-Fi (802.11), MQTT
- **Healthcare Standards & Interoperability:** HL7, IHE PCD, FHIR
- **AI & Developer Productivity Tools:** Large Language Models (LLMs), Prompt Engineering, GitHub Copilot
- **Software Engineering & System Design:** Object-Oriented Programming (OOP), Data Structures and Algorithms (DSA), Design Patterns, SOLID Principles, Agile (Scrum, Kanban)
- **Tools & Platforms:** Visual Studio, Visual Studio Code, GitHub, Windchill, SonarQube, Microsoft Excel
- **Operating Systems:** Windows, Linux
- **Leadership & Collaboration:** Team Leadership, Mentoring, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Resideo Smart Home Technologies Pvt. Ltd., Bengaluru

Advance Engineer I | Aug 2024 – Present

- Developed and maintained **tools bridging smart home IoT devices and application interfaces**, eliminating delays and dependency on the app during development and testing and reducing overall deliverable by **20%-25%**.
- Collaborated with firmware and application teams to ensure seamless **integration between tools and IoT devices**.

Philips India Ltd., Bengaluru

Software Technologist I | Apr 2024 – July 2024

- Managed **multiple program releases simultaneously**, including Serial drivers (EC40, EC10) and IHE Driver (LAN) development.
- Served as **Customer Escalations Lead**, resolving **41 out of 48 escalations** to date.
- I participated in **code reviews, sprint planning, project planning, and mentoring**, acting as the go-to person for technical blockers.

Software Engineer II | Oct 2021 – Mar 2024

- Owned and developed a **key integration feature** connecting BBraun Doselink+ solution with PICiX using **socket programming and networking protocols**.

- Managed **NLS IDs creation**, generating 300+ IDs per quarter to support team operations.
- Participated in **customer escalations, team hackathons, and cross-functional initiatives** to improve processes and collaboration.
- Implemented solutions using **C#, .NET, and SQL Server** to ensure reliable system integration and data management.

Software Engineer I | Jul 2019 – Sep 2021

- Developed and maintained **EC40 and EC10 embedded drivers** for medical devices, handling end-to-end development from requirements gathering to release and simulator creation.
- Delivered **15–18 drivers** over multiple years for devices including infusion systems, patient monitors, anesthesia machines, ventilators, and urine output systems.
- Implemented solutions using **C++, C#, and ACE Framework**, ensuring reliable device integration and performance.

Software Engineer Intern | Jan 2019 – Jul 2019

- Developed the **“E-Task” module** to allow clinicians in Remote Monitoring Centers to quickly record key patient notes.

PROJECTS

Smart Home IoT Tools Development

Wi-Fi IOP Framework Tool | Jan 2026 – Present

- **Objective:** Design a scalable test framework to validate IoT device compatibility across 100+ consumer routers.
- **Technical Highlights:** Java, Maven, BDD test automation, Wi-Fi interoperability testing. Used Network protocol to automate the verification of device connection stability across different 802.11 standards.
- **Impact:** Reduced manual IOP testing effort by 40% and Improved IoT device reliability across diverse network environments, identified router-specific issues early, and reduced field failures related to network incompatibility.

Matter Virtual Node Simulator | Sept 2025 – Dec 2025

- **Objective:** Developed a high-fidelity simulator to emulate Matter-enabled IoT devices (Lights, Thermostats, Sensors) to enable app testing without physical hardware.
- **Technical Highlights:** Utilized **Python** for tools development and **C++** to implement the Matter protocol stack over **IPv6/Thread**. Designed a state machine to handle Commissioning, Operational Discovery, and Cluster attribute reporting using MQTT.
- **Impact:** Eliminated hardware procurement delays for the QA team, accelerating the release cycle by 4 weeks

BLE Manager Tool | Dec 2024 – Aug 2025

- **Objective:** Simplify Bluetooth Low Energy (BLE) device discovery, secure connection, configuration, and cloud registration through a unified desktop tool.
- **Technical Highlights:** Python, PyQt5 GUI, BLE communication using bleak, secure BLE pairing, bidirectional data exchange, device authentication and configuration, ISU validation and summary, cross-platform support (Windows, macOS, Linux).
- **Impact:** Streamlined BLE device management, reduced pairing and communication complexity, and improved reliability of device onboarding and validation workflows.

Network Manager Tool | Aug 2024 – Nov 2024

- **Objective:** Build a Windows-based simulator to validate and test Network Manager functionality during early development, enabling end-to-end feature verification before UI availability.
- **Technical Highlights:** C#, .NET Framework 4.8, Windows Forms, RS232 serial communication, MQTT IPC, network scanning and configuration, connectivity monitoring.
- **Impact:** Enabled early functional testing, reduced integration risk, and accelerated Network Manager.

IntelliBridge Systems – Medical Device Integration (MDI)

IHE Driver (LAN) – PICiX (Patient Information Center iX) | Oct 2021 – Mar 2024

- **Objective:** Developed a standardized LAN driver to integrate the BBraun Doselink+ solution with the Philips Patient Information Center (PICiX).
- **Technical Highlights:** Optimized **Socket Programming (TCP/IP)** in **C#** to handle high-frequency vitals data. Ensured compliance with **IHE PCD (Patient Care Device)** profiles and used **HL7** for data formatting.
- **Impact:** Successfully integrated a major third-party infusion system into the Philips ecosystem, supporting critical care data flow for multiple hospital sites.

EC10 and EC40 Embedded Drivers | July 2019 – Sept 2021

- **Objective:** End-to-end development of serial communication drivers for diverse medical equipment (Ventilators, Anesthesia machines).
- **Technical Highlights:** Used **C++** and the **ACE Framework** for asynchronous I/O and event handling. Developed custom simulators to mimic serial data streams for 15+ different device types during the development phase.
- **Impact:** Standardized the driver development process, enabling the rapid onboarding of new hardware vendors within the Philips IntelliBridge platform.

EDUCATION

- MCA, National Institute of Technology Karnataka, Surathkal (2016–2019) – CGPA: 9.0/10
- BCA, University of Allahabad (2013–2016) – Percentage: 75%

ACHIEVEMENTS & LEADERSHIP

- Recognized for Philips values: Customer First, Take Ownership, Inspire & Improve
- Consistently rated 'Exceeds Expectations' in annual performance reviews
- Developed nimcet.in, used by multiple NITs for national entrance exams
- Secured AIR-120 in NIMCET 2016 (National MCA Entrance)
- Served as Event Head for IGNITE tech fest at NIT Surathkal
- Led customer escalations and NLS creation at Philips

LANGUAGES & INTERESTS

- Languages: English, Hindi
- Interests: Badminton, Cricket